

Performance and scalability report

Measured evidence

Measurement	Result	Interpretation
Staging HTML	865 bytes, 200, ~0.057 s curl	Edge response is fast; not representative of app readiness
Staging JS transfer	1,565,365 bytes compressed; local build 1,532.6 KiB gzip	High for Indian mobile data
Staging CSS transfer	39,047 bytes compressed	Moderate
Lighthouse mobile performance	34/100	Poor
Lighthouse FCP/LCP	12.5 s / 15.7 s	Launch blocker for low-bandwidth users
Lighthouse TBT/CLS	360 ms / 0.262	Interactivity/layout need work
Lighthouse total transfer	10,036 KiB	Includes images/other requests
Local JS/CSS output	6,114 KB / 245 KB minified	Vite warns chunks >500 KB
360px viewport	horizontal overflow confirmed	Budget Android/mobile ergonomics defect
Attached simulation	75 -> 375 req/s reported after query fix	Not independently reproducible; raw harness absent

Scale model (non-disruptive)

Load	Likely behavior	Evidence status
100 concurrent	Could fit a single staging task if cache/query fix holds	Hypothesis; no cloud test
1,000 concurrent	Rate-limit identity, single Redis, DB connections and external calls become risks	Not tested
10,000 concurrent	Max 10 tasks, DB/WS/event jobs/vendor quotas become constraints	Not demonstrated
100K-1M MAU	Requires cost/partitioning/job/analytics redesign and verified SLOs	Missing evidence

Main performance risks and corrections

1. Split admin, dashboards, chat, maps, voice and onboarding with route-level lazy/Suspense; defer providers and analytics.
2. Use responsive WebP/AVIF variants, srcset/sizes, dimensions and lazy loading below the fold.
3. Add RUM by network class/device and enforce JS/LCP/CLS budgets in CI.
4. Remove repeated full listing JSON from LLM context and reduce calls/schema surface.
5. Move rollups/supply jobs to singleton scheduled workers with leases.
6. Run authenticated k6 tests only in a dedicated isolated environment after service containers and limits are configured; never use production.

Findings

ID	Domain	Title	Severity	Confidence	Evidence	User/business impact	Recommended correction	Effort	Owner	Release blocker	Status
PERF-001	Web	LCP 15.7s and 10 MB transfer on mobile	High	Confirmed	Lighthouse 2026-07-10	Abandonment/data cost	Route split/media budgets	Medium	Frontend	yes	Open

ID	Domain	Title	Severity	Confidence	Evidence	User/business impact	Recommended correction	Effort	Owner	Release blocker	Status
PERF-002	Web	Monolithic 6.1 MB JS and eager routes	High	Confirmed	Vite build/App imports	Slow parse/memory	Dynamic imports and chunk budget	Medium	Frontend	yes	Open
PERF-003	Responsive	360px horizontal overflow	Medium	Confirmed	in-app browser viewport	Broken navigation/scroll	Fix overflow at 320/360/412px	Low/Medium	UX/Web	before beta	Open
PERF-004	Images	Lazy/responsive image discipline incomplete	Medium	Confirmed	source inspection	Bytes/memory/LCP	Variants/lazy/dimensions	Medium	Frontend/Media	no	Open
PERF-005	Scale evidence	375 req/s simulation cannot be reproduced	High	Confirmed missing artifact	PDF cites absent scratchpad	Investor/launch capacity claim unverified	Commit sanitized harness and rerun	Medium	Performance	yes	Open
PERF-006	External calls	AI/Maps/OTP/voice usage can scale cost faster than traffic	High	High confidence	provider code/PDF	Denial-of-wallet/outage	quotas, budgets, circuit breakers	Medium	SRE/CFO	yes	Open